

ArXiv Scalar vs AMC Tensor-Setup Equations (Term-by-Term)

Left column: scalar equations from `learn/arxiv_eq.tex` (J-coupled factorized section).

Right column: AMC-generated equations from `amc/examples/sample_output`.

Note: for equation 2 and equation 6, the requested tensor-typed intermediates (χ^β and χ^ζ) are shown directly from AMC output. The full scalar-final closure for those two terms fails in AMC with inconsistent projection/canonicalization errors under those tensor flags.

One-body I (AMC factorized: χ^α + closure)

ArXiv scalar

$$f_{ij}^I = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abJ} \hat{j}^2 \chi_{ab}^\alpha \Gamma_{biaj}^J$$

$$\chi_{ij}^\alpha = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab} \Omega_{abcj}$$

AMC generated (tensor setup)

$$\chi_{ij}^{\alpha 0} = \frac{1}{2} \cdot \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0J_1\lambda_0} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c \cdot n_i - n_a \cdot n_b \cdot \bar{n}_c \cdot \bar{n}_i) \cdot (-1)^{J_0+J_1+\lambda_0} \cdot \lambda_0^{-1} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Omega_{abcj}^{J_1J_0\lambda_0}$$

$$f_{ij}^{(I)0} =$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abJ_0} \cdot \delta_{j_b, j_a} \cdot \hat{j}_0^2 \cdot \chi_{ab}^{\alpha 0} \cdot \Gamma_{biaj}^{J_0J_00}$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abJ_0} \cdot \delta_{j_b, j_a} \cdot \hat{j}_0^2 \cdot \chi_{ab}^{\alpha 0} \cdot \Gamma_{aibj}^{J_0J_00}$$

One-body II

ArXiv scalar

$$f_{ij}^{II} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abJ} \hat{j}^2 (\chi_{ab}^\beta - \chi_{ba}^\beta) \Omega_{biaj}^J$$

$$\chi_{ij}^\beta = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c n_i - n_a n_b \bar{n}_c \bar{n}_i) \Omega_{ciab} \Gamma_{abcj}$$

AMC generated (tensor setup)

$$\chi_{ij}^{\beta \lambda_0} = \frac{1}{2} \cdot (-1)^{j_j + \lambda_0} \cdot \sum_{0abcJ_0J_1} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c \cdot n_i - n_a \cdot n_b \cdot \bar{n}_c \cdot \bar{n}_i) \cdot (-1)^{J_0+J_1} \cdot \hat{j}_0 \cdot \hat{j}_1 \cdot \left\{ \begin{matrix} J_0 & J_1 & \lambda_0 \\ j_j & j_i & j_c \end{matrix} \right\} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Gamma_{abcj}^{J_1J_10}$$

One-body IIIa

ArXiv scalar

$$f_{ij}^{IIIa} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abcJ} \left(\bar{\chi}_{i\bar{c}a\bar{b}}^{\gamma J} \Gamma_{abj\bar{c}}^J - \bar{\chi}_{c\bar{j}a\bar{b}}^{\gamma J} \Gamma_{ab\bar{c}\bar{i}}^J \right)$$

$$\chi_{ijkl}^{\gamma} = \sum_{ab} (n_a \bar{n}_b n_j \bar{n}_k - \bar{n}_a n_b \bar{n}_j n_k) \Omega_{ajkb} \Omega_{ibal}$$

AMC generated (tensor setup)

$$\chi_{ijkl}^{\gamma J_0 J_1 0} = -\delta_{J_1, J_0} \cdot (-1)^{j_j + j_l} \cdot \sum_{abJ_2 J_3 J_4 J_5 J_6 j_0 \lambda_0} \cdot (n_a \cdot \bar{n}_b \cdot n_j \cdot \bar{n}_k - \bar{n}_a \cdot n_b \cdot \bar{n}_j \cdot n_k) \cdot (-1)^{J_3 + J_5 + \lambda_0} \cdot \hat{J}_2 \cdot \hat{J}_3 \cdot \hat{J}_4 \cdot \hat{J}_5 \cdot \hat{J}_6^2 \cdot \hat{j}_0^2 \cdot \lambda_0^{\hat{1}} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_j & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_l & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_j & j_k & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_i & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_6 \end{matrix} \right\} \cdot \Omega_{ajkb}^{J_2 J_3 \lambda_0} \cdot \Omega_{ibal}^{J_4 J_5 \lambda_0}$$

$$f_{ij}^{(IIIa)^0} =$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{J}_0^2 \cdot \chi_{icab}^{\gamma J_0 J_0 0} \cdot \Gamma_{abjc}^{J_0 J_0 0}$$

$$- \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{J}_0^2 \cdot \Gamma_{ciab}^{J_0 J_0 0} \cdot \chi_{abcj}^{\gamma J_0 J_0 0}$$

One-body IIIb

ArXiv scalar

$$f_{ij}^{IIIb} = \delta_{j_i j_j} \hat{j}_i^{-2} \sum_{abcJ} (\chi_{ciab}^{\delta J} \Gamma_{abcj}^J - \chi_{abcj}^{\delta J} \Gamma_{ciab}^J)$$

$$\chi_{ijkl}^{\delta} = \frac{1}{4} \sum_{ab} (\bar{n}_i \bar{n}_j n_a n_b - n_i n_j \bar{n}_a \bar{n}_b) \Omega_{ijab} \Omega_{abkl}$$

AMC generated (tensor setup)

$$\chi_{ijkl}^{\delta J_0 J_1 0} = \frac{1}{4} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot \hat{J}_0^{-2} \cdot \sum_{abJ_2 \lambda_0} \cdot (\bar{n}_i \cdot \bar{n}_j \cdot n_a \cdot n_b - n_i \cdot n_j \cdot \bar{n}_a \cdot \bar{n}_b) \cdot (-1)^{J_2 + \lambda_0} \cdot \lambda_0^{\hat{1}} \cdot \Omega_{ijab}^{J_0 J_2 \lambda_0} \cdot \Omega_{abkl}^{J_2 J_0 \lambda_0}$$

$$f_{ij}^{(IIIb)^0} =$$

$$\delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{J}_0^2 \cdot \chi_{ciab}^{\delta J_0 J_0 0} \cdot \Gamma_{abcj}^{J_0 J_0 0}$$

$$- \delta_{j_j, j_i} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0} \cdot \hat{J}_0^2 \cdot \Gamma_{ciab}^{J_0 J_0 0} \cdot \chi_{abcj}^{\delta J_0 J_0 0}$$

Two-body I

ArXiv scalar

$$\Gamma_{ijkl}^{I,J} = \sum_a \left[(1 - \hat{P}_{ij}^J) \chi_{ai}^\epsilon \Gamma_{ajkl}^J + (1 - \hat{P}_{kl}^J) \chi_{ak}^\epsilon \Gamma_{ijal}^J \right]$$

$$\chi_{ij}^\epsilon = \frac{1}{2} \sum_{abc} (\bar{n}_a \bar{n}_b n_c + n_a n_b \bar{n}_c) \Omega_{ciab} \Omega_{abcj}$$

AMC generated (tensor setup)

$$\chi_{ij}^{\epsilon \ 0} = \frac{1}{2} \cdot \delta_{jj,ji} \cdot \hat{j}_i^{-2} \cdot \sum_{abcJ_0J_1\lambda_0} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_c + n_a \cdot n_b \cdot \bar{n}_c) \cdot (-1)^{J_0+J_1+\lambda_0} \cdot \hat{\lambda}_0^{-1} \cdot \Omega_{ciab}^{J_0J_1\lambda_0} \cdot \Omega_{abcj}^{J_1J_0\lambda_0}$$

$$\Gamma_{ijkl}^{(I) \ J_0J_10} =$$

$$\delta_{J_1,J_0} \cdot \sum_a \cdot \delta_{ja,ji} \cdot \chi_{ia}^{\epsilon \ 0} \cdot \Gamma_{ajkl}^{J_0J_10}$$

$$-\delta_{J_1,J_0} \cdot (-1)^{J_0+j_i+j_j} \cdot \sum_a \cdot \delta_{ja,jj} \cdot \chi_{ja}^{\epsilon \ 0} \cdot \Gamma_{aikl}^{J_0J_10}$$

$$\delta_{J_1,J_0} \cdot \sum_a \cdot \delta_{ja,jk} \cdot \chi_{ak}^{\epsilon \ 0} \cdot \Gamma_{ijal}^{J_0J_10}$$

$$-\delta_{J_1,J_0} \cdot (-1)^{J_0+j_k+j_l} \cdot \sum_a \cdot \delta_{ja,jl} \cdot \chi_{al}^{\epsilon \ 0} \cdot \Gamma_{ijak}^{J_0J_10}$$

Two-body II

ArXiv scalar

$$\Gamma_{ijkl}^{II,J} = \sum_a \left[(1 - \hat{P}_{ij}^J) \chi_{aj}^\zeta \Omega_{iakl}^J - (1 - \hat{P}_{kl}^J) \chi_{ak}^\zeta \Omega_{ijal}^J \right]$$

$$\chi_{ij}^\zeta = \frac{1}{2} \sum_{abc} (n_a n_b \bar{n}_c + \bar{n}_a \bar{n}_b n_c) \Gamma_{aibc} \Omega_{bcaj}$$

AMC generated (tensor setup)

$$\chi_{ij}^{\zeta \ \lambda_0} = \frac{1}{2} \cdot (-1)^{j_j+\lambda_0} \cdot \sum_{0abcJ_0J_1} \cdot (n_a \cdot n_b \cdot \bar{n}_c + \bar{n}_a \cdot \bar{n}_b \cdot n_c) \cdot (-1)^{J_0+j_a} \cdot \hat{j}_0 \cdot \hat{j}_1 \cdot \left\{ \begin{matrix} \lambda_0 & J_1 & J_0 \\ j_a & j_i & j_j \end{matrix} \right\} \cdot \Gamma_{aibc}^{J_0J_10} \cdot \Omega_{bcaj}^{J_0J_1\lambda_0}$$

Two-body IIIa

ArXiv scalar

$$\Gamma_{ijkl}^{IIIa,J} = \sum_{ab} \left[(1 - \hat{P}_{ij}^J) \chi_{ijab}^{\eta J} \Gamma_{abkl}^J + (1 - \hat{P}_{kl}^J) \Gamma_{ijab}^J \chi_{abkl}^{\eta J} \right]$$

$$\chi_{ijkl}^{\eta} = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{iabl} \Omega_{bjka}$$

AMC generated (tensor setup)

$$\chi_{ijkl}^{\eta J_0 J_1 0} = -\delta_{J_1, J_0} \cdot (-1)^{j_i + j_k} \cdot \sum_{ab J_2 J_3 J_4 J_5 J_6 j_0 \lambda_0} \cdot (\bar{n}_a \cdot n_b \cdot \bar{n}_k + n_a \cdot \bar{n}_b \cdot n_k) \cdot (-1)^{J_2 + J_4 + \lambda_0} \cdot \hat{J}_2 \cdot \hat{J}_3 \cdot \hat{J}_4 \cdot \hat{J}_5 \cdot \hat{J}_6^2 \cdot \hat{j}_0^2 \cdot \lambda_0^{-1} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_i & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_k & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_6 \end{matrix} \right\} \cdot \Omega_{iabl}^{J_2 J_3 \lambda_0} \cdot \Omega_{bjka}^{J_4 J_5 \lambda_0}$$

$$\Gamma_{ijkl}^{(IIIa) J_0 J_1 0} =$$

$$\delta_{J_1, J_0} \cdot \sum_{ab} \cdot \chi_{ijab}^{\eta J_0 J_0 0} \cdot \Gamma_{abkl}^{J_0 J_0 0}$$

$$-\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j} \cdot \sum_{ab} \cdot \chi_{jiab}^{\eta J_0 J_0 0} \cdot \Gamma_{abkl}^{J_0 J_0 0}$$

$$\delta_{J_1, J_0} \cdot \sum_{ab} \cdot \Gamma_{ijab}^{J_0 J_0 0} \cdot \chi_{abkl}^{\eta J_0 J_0 0}$$

$$-\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_k + j_l} \cdot \sum_{ab} \cdot \Gamma_{ijab}^{J_0 J_0 0} \cdot \chi_{ablk}^{\eta J_0 J_0 0}$$

Two-body IIIb

ArXiv scalar

$$\bar{\bar{\Gamma}}_{j\bar{l}k\bar{i}}^{III_b, J} = (1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\bar{\Gamma}}_{j\bar{l}a\bar{b}}^J \left(\bar{\bar{\chi}}_{a\bar{b}k\bar{i}}^{\eta J} + \bar{\bar{\chi}}_{k\bar{i}b\bar{a}}^{\eta J} \right)$$

$$\chi_{ijkl}^{\eta} = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{iabl} \Omega_{bjka}$$

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^{\eta J_0 J_1 0} = & -\delta_{J_1, J_0} \cdot (-1)^{j_i + j_k} \cdot \sum_{ab J_2 J_3 J_4 J_5 J_6 j_0 \lambda_0} \\ & \cdot (\bar{n}_a \cdot n_b \cdot \bar{n}_k + n_a \cdot \bar{n}_b \cdot n_k) \cdot (-1)^{J_2 + J_4 + \lambda_0} \\ & \cdot \hat{J}_2 \cdot \hat{J}_3 \cdot \hat{J}_4 \cdot \hat{J}_5 \cdot \hat{J}_6^2 \cdot \hat{j}_0^2 \cdot \lambda_0^{-1} \cdot \left\{ \begin{matrix} J_3 & \lambda_0 & J_2 \\ j_a & j_i & j_0 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} J_4 & \lambda_0 & J_5 \\ j_a & j_k & j_0 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_6 \\ j_b & j_0 & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_6 \\ j_b & j_0 & J_4 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_6 \end{matrix} \right\} \cdot \Omega_{iabl}^{J_2 J_3 \lambda_0} \cdot \Omega_{bjka}^{J_4 J_5 \lambda_0} \end{aligned}$$

$$\Gamma_{ijkl}^{(III_b) J_0 J_1 0} =$$

$$\begin{aligned} & -\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \\ & \cdot (-1)^{J_2 + J_3 + j_a + j_b} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_i & j_k & J_4 \\ j_b & j_a & J_2 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_j & J_4 \\ j_b & j_a & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{bkai}^{\eta J_2 J_2 0} \cdot \Gamma_{jbla}^{J_3 J_3 0} \end{aligned}$$

$$\begin{aligned} & -\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \\ & \cdot (-1)^{J_2 + J_3 + j_a + j_b} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_j & j_l & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_k & j_i & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{lajb}^{\eta J_2 J_2 0} \cdot \Gamma_{aibk}^{J_3 J_3 0} \end{aligned}$$

$$\begin{aligned} & \delta_{J_1, J_0} \cdot (-1)^{j_i + j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_i & j_l & J_4 \\ j_b & j_a & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_4 \\ j_b & j_a & J_3 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{blai}^{\eta J_2 J_2 0} \cdot \Gamma_{jbka}^{J_3 J_3 0} \end{aligned}$$

$$\begin{aligned} & \delta_{J_1, J_0} \cdot (-1)^{j_i + j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_j & j_k & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_i & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{kajb}^{\eta J_2 J_2 0} \cdot \Gamma_{aibl}^{J_3 J_3 0} \end{aligned}$$

$$\begin{aligned} & \delta_{J_1, J_0} \cdot (-1)^{j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_j & j_k & J_4 \\ j_b & j_a & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_i & J_4 \\ j_b & j_a & J_3 \end{matrix} \right\} \\ & \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{bkaj}^{\eta J_2 J_2 0} \cdot \Gamma_{ibla}^{J_3 J_3 0} \end{aligned}$$

$$\begin{aligned} & \delta_{J_1, J_0} \cdot (-1)^{j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\ & \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_i & j_l & J_4 \\ j_b & j_a & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_4 \\ j_b & j_a & J_3 \end{matrix} \right\} \end{aligned}$$

Two-body IIIc

ArXiv scalar

$$\bar{\Gamma}_{i\bar{l}k\bar{j}}^{III_c, J} = \frac{1}{2}(1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\chi}_{i\bar{l}a\bar{b}}^{\theta J} \bar{\Gamma}_{a\bar{b}k\bar{j}}^J$$

$$\chi_{ijkl}^{\theta} = \sum_{ab} (n_a n_b \bar{n}_k + \bar{n}_a \bar{n}_b n_k + n_a n_b \bar{n}_j + \bar{n}_a \bar{n}_b n_j) \Omega_{ijab} \Omega_{abkl}$$

AMC generated (tensor setup)

$$\chi_{ijkl}^{\theta J_0 J_1 0} = \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot \hat{J}_0^{-2} \cdot \sum_{ab J_2 \lambda_0} \cdot (n_a \cdot n_b \cdot \bar{n}_k + \bar{n}_a \cdot \bar{n}_b \cdot n_k + n_a \cdot n_b \cdot \bar{n}_j + \bar{n}_a \cdot \bar{n}_b \cdot n_j) \cdot (-1)^{J_2 + \lambda_0} \cdot \hat{\lambda}_0^{-1} \cdot \Omega_{ija\bar{b}}^{J_0 J_2 \lambda_0} \cdot \Omega_{abkl}^{J_2 J_0 \lambda_0}$$

$$\Gamma_{ijkl}^{(III_c) J_0 J_1 0} =$$

$$-\frac{1}{2} \cdot \delta_{J_1, J_0} \cdot \sum_{ab J_2 J_3 J_4} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_l & j_i & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_j & j_k & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{iabl}^{\theta J_2 J_2 0} \cdot \Gamma_{bjka}^{J_3 J_3 0}$$

$$\frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0 + j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_k & j_i & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_j & j_l & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{iabk}^{\theta J_2 J_2 0} \cdot \Gamma_{bjla}^{J_3 J_3 0}$$

$$\frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_l & j_j & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_k & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{jabl}^{\theta J_2 J_2 0} \cdot \Gamma_{bika}^{J_3 J_3 0}$$

$$-\frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot \hat{J}_2^2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_k & j_j & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{jabk}^{\theta J_2 J_2 0} \cdot \Gamma_{bila}^{J_3 J_3 0}$$

Two-body IVa

ArXiv scalar

$$\Gamma_{ijkl}^{IV_a, J} = - \sum_{ab} \left[(1 - \hat{P}_{ij}^J) \chi_{ijab}^{\kappa J} \Omega_{abkl}^J + (1 - \hat{P}_{kl}^J) \Omega_{ijab}^J \chi_{klab}^{\kappa J} \right]$$

$$\chi_{ijkl}^{\kappa} = \sum_{ab} (\bar{n}_a n_b n_k + n_a \bar{n}_b \bar{n}_k) \Omega_{ajbl} \Gamma_{ibka}$$

AMC generated (tensor setup)

$$\chi_{ijkl}^{\kappa J_0 J_1 0} = \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot \sum_{ab J_2 J_3 J_4} \cdot (\bar{n}_a \cdot n_b \cdot n_k + n_a \cdot \bar{n}_b \cdot \bar{n}_k)$$

$$\cdot (-1)^{J_2 + J_3 + j_a + j_b} \cdot \hat{J}_2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_l & j_j & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\}$$

$$\cdot \left\{ \begin{matrix} j_i & j_k & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \Omega_{ajbl}^{J_2 J_2 0} \cdot \Gamma_{ibka}^{J_3 J_3 0}$$

$$\Gamma_{ijkl}^{(IV_a) J_0 J_1 0} =$$

$$-\delta_{J_1, J_0} \cdot \hat{J}_0^{-1} \cdot \sum_{ab} \cdot \chi_{ijab}^{\kappa J_0 J_0 0} \cdot \Omega_{abkl}^{J_0 J_0 0}$$

$$\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j} \cdot \hat{J}_0^{-1} \cdot \sum_{ab} \cdot \chi_{jiab}^{\kappa J_0 J_0 0} \cdot \Omega_{abkl}^{J_0 J_0 0}$$

$$-\delta_{J_1, J_0} \cdot \hat{J}_0^{-1} \cdot \sum_{ab} \cdot \Omega_{ijab}^{J_0 J_0 0} \cdot \chi_{klab}^{\kappa J_0 J_0 0}$$

$$\delta_{J_1, J_0} \cdot (-1)^{J_0 + j_k + j_l} \cdot \hat{J}_0^{-1} \cdot \sum_{ab} \cdot \Omega_{ijab}^{J_0 J_0 0} \cdot \chi_{lkab}^{\kappa J_0 J_0 0}$$

Two-body IVb

ArXiv scalar

$$\bar{\bar{\Gamma}}_{j\bar{l}k\bar{i}}^{IV_b, J} = (1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\bar{\Omega}}_{j\bar{l}a\bar{b}}^J (\bar{\chi}_{a\bar{b}k\bar{i}}^J - \bar{\chi}_{k\bar{i}a\bar{b}}^J)$$

$$\chi_{ijkl}^\iota = \sum_{ab} (\bar{n}_a n_b \bar{n}_k + n_a \bar{n}_b n_k) \Omega_{bika} \Gamma_{jabl}$$

AMC generated (tensor setup)

$$\begin{aligned} \chi_{ijkl}^\iota J_0 J_1^0 &= \delta_{J_1, J_0} \cdot (-1)^{J_0+j_i+j_j} \cdot \sum_{ab J_2 J_3 J_4} \\ &\cdot (\bar{n}_a \cdot n_b \cdot \bar{n}_k + n_a \cdot \bar{n}_b \cdot n_k) \cdot \hat{J}_2 \cdot \hat{J}_3^2 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_k & j_i & J_4 \\ j_b & j_a & J_2 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_j & j_l & J_4 \\ j_b & j_a & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \Omega_{bika}^{J_2 J_2^0} \cdot \Gamma_{jabl}^{J_3 J_3^0} \end{aligned}$$

$$\Gamma_{ijkl}^{(IV_b)} J_0 J_1^0 =$$

$$\begin{aligned} &\delta_{J_1, J_0} \cdot (-1)^{J_0+j_i+j_j+j_k+j_l} \cdot \sum_{ab J_2 J_3 J_4} \\ &\cdot (-1)^{J_2+J_3+j_a+j_b} \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_k & j_i & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_j & j_l & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{aibk}^\iota J_2 J_2^0 \cdot \Omega_{jbla}^{J_3 J_3^0} \end{aligned}$$

$$\begin{aligned} &-\delta_{J_1, J_0} \cdot (-1)^{J_0+j_i+j_j+j_k+j_l} \cdot \sum_{ab J_2 J_3 J_4} \\ &\cdot (-1)^{J_2+J_3+j_a+j_b} \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_i & j_k & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_l & j_j & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{akbi}^\iota J_2 J_2^0 \cdot \Omega_{jalb}^{J_3 J_3^0} \end{aligned}$$

$$\begin{aligned} &-\delta_{J_1, J_0} \cdot (-1)^{j_i+j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2+J_3+j_a+j_b} \\ &\cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_l & j_i & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_j & j_k & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{aibl}^\iota J_2 J_2^0 \cdot \Omega_{jbka}^{J_3 J_3^0} \end{aligned}$$

$$\begin{aligned} &\delta_{J_1, J_0} \cdot (-1)^{j_i+j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2+J_3+j_a+j_b} \\ &\cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_i & j_l & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_k & j_j & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_l & j_k & J_0 \\ j_j & j_i & J_4 \end{matrix} \right\} \cdot \chi_{albi}^\iota J_2 J_2^0 \cdot \Omega_{jabk}^{J_3 J_3^0} \end{aligned}$$

$$\begin{aligned} &-\delta_{J_1, J_0} \cdot (-1)^{j_k+j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2+J_3+j_a+j_b} \\ &\cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_k & j_j & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_i & j_l & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{ajbk}^\iota J_2 J_2^0 \cdot \Omega_{ibla}^{J_3 J_3^0} \end{aligned}$$

$$\begin{aligned} &\delta_{J_1, J_0} \cdot (-1)^{j_k+j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2+J_3+j_a+j_b} \\ &\cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \left\{ \begin{matrix} j_j & j_k & J_4 \\ j_a & j_b & J_2 \end{matrix} \right\} \cdot \left\{ \begin{matrix} j_l & j_i & J_4 \\ j_a & j_b & J_3 \end{matrix} \right\} \\ &\cdot \left\{ \begin{matrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{matrix} \right\} \cdot \chi_{akbj}^\iota J_2 J_2^0 \cdot \Omega_{ialb}^{J_3 J_3^0} \end{aligned}$$

Two-body IVc

ArXiv scalar

AMC generated (tensor setup)

$$\begin{aligned}
\bar{\Gamma}_{i\bar{l}k\bar{j}}^{IVc,J} &= \frac{1}{2}(1 - \hat{P}_{ij}^J)(1 - \hat{P}_{kl}^J) \sum_{ab} \bar{\chi}_{i\bar{l}a\bar{b}}^{\lambda J} \bar{\Omega}_{a\bar{k}j\bar{b}}^J \\
\chi_{ijkl}^\lambda &= \sum_{ab} (\bar{n}_a \bar{n}_b n_l + n_a n_b \bar{n}_l) \Gamma_{ijab} \Omega_{abkl} + \sum_{ab} (\bar{n}_a \bar{n}_b n_j + n_a n_b \bar{n}_j) \Omega_{ijab} \bar{\Gamma}_{abkl}^{\hat{c}} \sum_{ab} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_l + n_a \cdot n_b \cdot \bar{n}_l) \cdot \Gamma_{ijab}^{J_0 J_0 0} \cdot \Omega_{abkl}^{J_0 J_0 0} \\
&\quad \delta_{J_1, J_0} \cdot J_0^{-1} \cdot \sum_{ab} \cdot (\bar{n}_a \cdot \bar{n}_b \cdot n_j + n_a \cdot n_b \cdot \bar{n}_j) \cdot \Omega_{ijab}^{J_0 J_0 0} \cdot \Gamma_{abkl}^{J_0 J_0 0} \\
\Gamma_{ijkl}^{(IVc)}^{J_0 J_1 0} &= \\
&\quad \frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\
&\quad \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \begin{Bmatrix} j_l & j_i & J_4 \\ j_a & j_b & J_2 \end{Bmatrix} \cdot \begin{Bmatrix} j_j & j_k & J_4 \\ j_a & j_b & J_3 \end{Bmatrix} \\
&\quad \cdot \begin{Bmatrix} j_l & j_k & J_0 \\ j_j & j_i & J_4 \end{Bmatrix} \cdot \chi_{ialb}^{\lambda J_2 J_2 0} \cdot \Omega_{bjak}^{J_3 J_3 0} \\
&\quad - \frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\
&\quad \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \begin{Bmatrix} j_k & j_i & J_4 \\ j_a & j_b & J_2 \end{Bmatrix} \cdot \begin{Bmatrix} j_j & j_l & J_4 \\ j_a & j_b & J_3 \end{Bmatrix} \\
&\quad \cdot \begin{Bmatrix} j_k & j_l & J_0 \\ j_j & j_i & J_4 \end{Bmatrix} \cdot \chi_{iakb}^{\lambda J_2 J_2 0} \cdot \Omega_{bjal}^{J_3 J_3 0} \\
&\quad - \frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{J_0 + j_i + j_j + j_k + j_l} \cdot \sum_{ab J_2 J_3 J_4} \\
&\quad \cdot (-1)^{J_2 + J_3 + j_a + j_b} \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \begin{Bmatrix} j_l & j_j & J_4 \\ j_a & j_b & J_2 \end{Bmatrix} \\
&\quad \cdot \begin{Bmatrix} j_i & j_k & J_4 \\ j_a & j_b & J_3 \end{Bmatrix} \cdot \begin{Bmatrix} j_l & j_k & J_0 \\ j_i & j_j & J_4 \end{Bmatrix} \cdot \chi_{jalb}^{\lambda J_2 J_2 0} \cdot \Omega_{biak}^{J_3 J_3 0} \\
&\quad \frac{1}{2} \cdot \delta_{J_1, J_0} \cdot (-1)^{j_i + j_j} \cdot \sum_{ab J_2 J_3 J_4} \cdot (-1)^{J_2 + J_3 + j_a + j_b} \\
&\quad \cdot \hat{J}_2^2 \cdot \hat{J}_3 \cdot \hat{J}_4^2 \cdot \begin{Bmatrix} j_k & j_j & J_4 \\ j_a & j_b & J_2 \end{Bmatrix} \cdot \begin{Bmatrix} j_i & j_l & J_4 \\ j_a & j_b & J_3 \end{Bmatrix} \\
&\quad \cdot \begin{Bmatrix} j_k & j_l & J_0 \\ j_i & j_j & J_4 \end{Bmatrix} \cdot \chi_{jakb}^{\lambda J_2 J_2 0} \cdot \Omega_{bial}^{J_3 J_3 0}
\end{aligned}$$